

UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program / Fourth Semester – 2021

Roll No. 0335

Paper: Statistics-IV
Course Code: STAT-203 / STT-22314 Part - II

Time: 2 Hrs. 45 Min. Marks: 60

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Solve the following:

(4x5=20)

- Define a chi-square random variable and its density function.
- Define a contingency table. How do you determine the number of degree of freedom in an $(r \times c)$ contingency table?
- How will you find a confidence interval for population correlation coefficients?
- Describe Latin square design and its analysis. How does it differ from randomized complete block design?

Solve the following:

Q # 03

(30)

- Employers want to know which days of the week employees are absent in a five-day work week. Most employers would like to believe that employees are absent equally during the week. Suppose a random sample of 60 managers were asked on which day of the week they had the highest number of employee absences. The results were distributed as in the table below. For the population of employees, do the days for the highest number of absences occur with equal frequencies during a five-day work week? Test at a 5% significance level. Day of the Week Employees were Most Absent:

	Monday	Tuesday	Wednesday	Thursday	Friday
Number of Absences	15	12	9	9	15

- An education researcher is comparing four different algebra curricula. Eighth grade students are randomly assigned to one-one of the four groups. Their state achievement test scores are compared at the end of the year. Use the appropriate statistical procedure to determine whether the curricula differ with respect to math achievement. An alpha criterion of 0.05 should be used for the test.

	n	mean	sd
curriculum 1	50	170.5	14.6
curriculum 2	50	168.3	12.8
curriculum 3	50	167.6	17.7
curriculum 4	50	172.8	16.8

- Given the data,

Fertilizer (X)	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4
Corn Yield (Y)	10	15	30	35	25	30	50	45

Assuming normality, calculate the 95% confidence interval for

- The value of α
- The value of β
- The true value of Y when X = 3.0

2021

Last 10 days

question 2(1)

Q Define chi-square random variables and its density Function? only (0.5)

* The chi-square distribution contains one parameter called the number of degree of Freedom (d.f.).

Where the number of degree of freedom represent the number of independent random variables called chi square.

If a random variable entering a chi-square subjected to linear restrictions, then the no of degree of freedom is reduced by the number of restriction involved.

$$f(x^2) = \frac{1}{2^{n/2} \Gamma(n/2)} (x^2)^{(n/2)-1} e^{-x^2/2} \quad 0 < x^2 < \infty$$

* The mean of chi square distribution is equal to the no of degree of freedom $E(x^2_{(n)}) = n$. Where the variance is equal to twice the number of degree of freedom.

2021 + 2018 $Var(x^2_{(n)}) = 2n$

17.7.1. Co-efficient of Contingency for an $r \times c$ contingency table. The chi-square statistic shows only whether the sample data do or do not conform to the hypothesis. It does not tell anything about the strength of the association, which we sometimes desire to measure. For this purpose, Karl Pearson (1857–1936) has defined a co-efficient C , known as the *Pearson's co-efficient of mean-square contingency*, by the relation

question2(2)

Population correlation coefficient = ρ_{xy}

μ_z is the parameter of ρ_{xy}

100(1- α)% C.I. for μ_z is

$$Z_f \pm \frac{Z_{\alpha/2}}{\sqrt{n-3}}$$

where

$$Z_f = 1.1513 \log \left(\frac{1+r}{1-r} \right)$$

As r is the sample correlation coefficient obtained from a random sample of n pairs of values from a bivariate normal population having a linear correlation ρ

r is used to estimate the value of ρ .

$$\mu_z = 1.1513 \log \left(\frac{1+\rho}{1-\rho} \right)$$

RCBD	Latin square
Suitable to heterogeneous material with respect to one character.	Suitable to heterogeneous material with respect to two characters.
Can be used safely up to 20 treatments	Not suitable for less than 5 and more than 12 treatments.
No. of replications is same for all treatments	No. of replication is equal to no. of treatments
All treatments are randomized in every replication by random number.	Treatments are as randomized as every treatment occurs once in each row and column.
Heterogeneous experimental material divided into groups of homogenous experimental unit called block/replication.	Experimental materials is divided into rows and columns based on two source of variation.

LATIN SQUARE DESIGN

In RBD whole of the experimental area is divided into relatively homogeneous groups (blocks) and treatments are allocated at random to units within each block, i.e., randomisation was subjected to one restriction i.e., within blocks. A useful method of eliminating fertility variations consists in an experimental layout which will control variation in two perpendicular directions. Such a layout is a Latin Square Design (LSD).

LAYOUT OF THE DESIGN

In this design the number of treatments is equal to the number of replications. Thus in the case of m treatments there have to be $m \times m = m^2$ experimental units. The whole of the experimental area is divided into m^2 experimental units (plots) arranged in a square so that each row as well as each column contains m units (plots). The m treatments are then allocated at random to these rows and columns in such a way that every treatment then occurs once and only once in each row and in each column. Such a layout is known as $m \times m$ Latin Square Design (LSD).

question 2(2)

$$C = \sqrt{\frac{\chi^2}{n + \chi^2}},$$

where n indicates sample size.

This co-efficient measures the strength of the association or dependence of two variables of classification of a contingency table. When there is complete independence, $C=0$. When the two classifications are perfectly associated, $C = \sqrt{(k-1)/k}$, where k is the smaller of r and c . Thus C lies between 0 and $\sqrt{(k-1)/k}$. In a 2×3 contingency table, the maximum value of C is $\sqrt{(2-1)/2} = 0.707$. The larger the value of C , the stronger is the association or dependence.

Another measure, known as *Cramer's co-efficient of contingency*, is defined as

significance possible. Each of them is described briefly in the following subsections.

question 2(4)

23.2.1. Randomization. (The first principle of an experimental design is *randomization*, which is a random process of assigning treatments to the experimental units.) The random process implies that every possible allotment of treatments has the same probability. An *experimental unit* is the smallest division of the experimental material and a *treatment* means an experimental condition whose effect is to be measured and compared. The purpose of randomization is to remove bias and other sources of extraneous variation, which are not controllable. Another advantage of randomization (accompanied by replication) is that it forms the basis of any valid statistical test. Hence the treatments must be assigned *at random* to the experimental units. Randomization is usually done by drawing numbered cards from a well-shuffled pack of cards, or by drawing numbered balls from a well-shaken container or by using tables of random numbers.

* The Least Significant Difference Tests:-

The Least ~~Square~~ test is used in the context of the analysis of Variance. When F-ratio suggest rejection of the null hypothesis H_0 that is difference between population mean is significant.

$$LSD = t_{\alpha/2}(v) \sqrt{\frac{2(MSE)}{r}}$$

question 22 (7)

2021 Employee want to know which days of the week employee
 Q no 3 (a) Present to a five-day work week. Must

	Monday	Tues	wed	Thur	Fri	Sat	Sun
No of Absence	15	12	9	9	15	15	15

① Hypothesis:-

$$H_0: P_1 = P_2 = P_3 = P_4 = P_5 = 1/5$$

$$H_1: P_i \neq P_{i0}$$

question no 3(a)

② L.O.S:- $\alpha = 0.05$

③ Test statistic:-

$$\chi^2 = \frac{(n_i - n_i p_{i0})^2}{n_i p_{i0}}$$

④ Calculations

	$f(n_i)$	$n p_{i0}$	
1	15	$60 \times \frac{1}{5} = 12$	$\frac{(15-12)^2}{12} = 0.75$
2	12	24 12	0
3	9	12	0.75
4	9	12	0.75
5	15	12	0.75
	60	12	3

question 3(a)

⑤ Critical region - We will reject H_0
 $\chi^2 \geq \chi_{0.05(5-1)}$
 $\chi^2 \geq 9.49$

⑥ Conclusion As calculated value of χ^2 does not lies in critical region so we will not reject H_0

2018

2021

2021 QNo3(b)

 $n_1 = n_2$

$$n_1 = n_2 = n_3 = n_4 = 50$$

$$\bar{X}_1 = 170.5$$

$$\bar{X}_2 = 168.3$$

$$\bar{X}_3 = 167.6$$

$$\bar{X}_4 = 172.8$$

$$s.d_1 = 14.5 \quad s.d_2 = 12.8$$

$$s.d_3 = 17.7$$

$$s.d_4 = 16.8$$

① Hypothesis:-

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

 H_1 : Not all mean are equal

② Level of Significance-

$$\alpha = 0.05$$

③ Test statistic:-

$$F = \frac{s_b^2}{s_w^2}$$

④ Calculations

$$\begin{aligned} \bar{X}_{..} &= \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2 + n_3 \bar{X}_3 + n_4 \bar{X}_4 + \cancel{n_5 \bar{X}_5}}{n_1 + n_2 + n_3 + n_4 + \cancel{n_5}} \\ &= \frac{50(170.5) + 50(168.3) + 50(167.6) + 50(172.8) + 50(1)}{50 + 50 + 50 + 50} \\ &= \frac{8525 + 8415 + 8380 + 8640}{200} = \frac{33960}{200} = 169.8 \end{aligned}$$

$$\text{Between S.S} = \sum_{j=1}^k n_j (\bar{X}_{.j} - \bar{X}_{..})^2$$

$$= (50(170.5 - 169.8)^2 + 50(168.3 - 169.8)^2 + 50(167.6 - 169.8)^2 + 50(172.8 - 169.8)^2)$$

$$= 24.5 + 112.5 + 72 + 450 = 659$$

$$\begin{aligned} \text{Within S.S} &= (n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + (n_3 - 1)s_3^2 + (n_4 - 1)s_4^2 \\ &= (50 - 1)(14.5)^2 + (50 - 1)(12.8)^2 + (50 - 1)(17.7)^2 + (50 - 1)(16.8)^2 \\ &= 10825 + 8028.5 + 15351.21 + 13829.71 \end{aligned}$$

$$\text{Inf} = 10825 + 8028.5 + 15351.21 + 13829.71$$

2018

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Within S.S = 140234.72

Source of Variation	df	Sum of Square	Mean Square	F
Between	$k-1 = 3$	4 659	219.67	$S_b^2 = 219.67$
Within	$n-k = 200 - 3 = 197$	200 140234.72	711.85	$S_w^2 = 711.85$
	200	140234.72		$= 0.21$

Critical region - We will reject H_0

$$F_c \geq F_{\alpha/2}(v_1, v_2)$$

$$0.21 > F_{0.025}(3, 197)$$

$$0.21 \nless 2.6$$

Conclusion - As calculated Value of F does not lies in critical region so we will not reject H_0 Accept H_0

QNO 3(c)

Example 21.2. Given the data (202)

Fertilizer (X)	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4
Yield (Y)	10	15	30	35	25	30	50	45

Assuming normality, calculate the 95% confidence intervals for (i) the value of α , (ii) the value of β , (iii) the true value of Y when $X=3.0$.

To find sample estimates of various parameters in the regression, the necessary computations are given below:

X	Y	XY	X ²	Y ²
0.3	10	3.0	0.09	100
0.6	15	9.0	0.36	225
0.9	30	27.0	0.81	900
1.2	35	42.0	1.44	1225
1.5	25	37.5	2.25	625
1.8	30	54.0	3.24	900
2.1	50	105.0	4.41	2500
2.4	45	108.0	5.76	2025
10.8	240	385.5	18.36	8500

question 3(c)

$$\text{Now } b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} = \frac{(8)(385.5) - (10.8)(240)}{(8)(18.36) - (10.8)^2}$$

$$= \frac{3084 - 2592}{146.88 - 116.64} = \frac{492}{30.24} = 16.27, \text{ and}$$

$$a = \bar{Y} - b\bar{X} = 30 - (16.27)(1.35) = 8.04.$$

The estimated regression line is

$$\hat{Y} = 8.04 + 16.27X.$$

Further,

$$\begin{aligned}\sum(Y - \hat{Y})^2 &= \sum Y^2 - a\sum Y - b\sum XY \\ &= 8500 - (8.04)(240) - (16.27)(385.5) \\ &= 8500 - 1929.6 - 6272.08 = 298.32, \text{ and} \\ s_{Y.X} &= \sqrt{\frac{\sum(Y - \hat{Y})^2}{n-2}} = \sqrt{\frac{298.32}{6}} = 7.05\end{aligned}$$

$$\text{And } \sum(X_i - \bar{X})^2 = \sum X^2 - (\sum X)^2/n = 18.36 - 14.56 = 3.78$$

(i) The 95% confidence interval for α is given by

$$a \pm t_{\alpha/2, (n-2)} s_a$$

Now $a = 8.04$, $t_{0.025, (6)} = 2.447$, and (from t -tables)

$$s_a = s_{Y.X} \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum(X - \bar{X})^2}}$$

question 3(c)

$$\begin{aligned}&= 7.05 \sqrt{\frac{1}{8} + \frac{(1.35)^2}{3.78}} = 7.05 \sqrt{0.125 + 0.482} \\ &= (7.05)(0.779) = 5.492\end{aligned}$$

Substituting these values, we get

$$8.04 \pm (2.447)(5.492)$$

$$\text{or } 8.04 \pm 13.44 \text{ or } -5.40 \text{ to } 21.48.$$

Hence the 95% confidence interval for α is $(-5.40, 21.48)$.

(ii) The 95% confidence interval for β is given by

$$b \pm t_{\alpha/2, (n-2)} s_b$$

Now $b = 16.27$, $t_{0.025, (6)} = 2.447$, and

$$s_b = \frac{s_{Y.X}}{\sqrt{\sum(X - \bar{X})^2}} = \frac{7.05}{\sqrt{3.78}} = \frac{7.05}{1.94} = 3.63$$

Substituting these values, we get

$$16.27 \pm (2.447)(3.63)$$

$$\text{or } 16.27 \pm 8.88, \text{ or } 7.39 \text{ to } 25.15.$$

Hence the 95% confidence interval for β is $(7.39, 25.15)$.

(iii) The 95% confidence interval for the true value of Y , i.e. $\mu_{Y,X}$ when $X = 3.0$, is given by

$$\hat{Y} \pm t_{\alpha/2, (n-2)} s_{\hat{Y}}$$

Now the estimated value of Y for $X=3.0$ is

$$\hat{Y} = a + bX_0 = 8.04 + (16.27)(3.0) = 56.85,$$

$$t_{0.025, (6)} = 2.447, \text{ and}$$

$$s_{\hat{Y}} = s_{Y.X} \sqrt{\frac{1}{n} + \frac{(\hat{X}_0 - \bar{X})^2}{\sum (X - \bar{X})^2}}$$

$$= 7.05 \sqrt{\frac{1}{8} + \frac{(3.0 - 1.35)^2}{3.78}}$$

$$(\because X_0 = 3.0)$$

$$= 7.05 \sqrt{0.125 + 0.7202}$$

$$= (7.05)(0.919) = 6.48$$

Substituting these values, we obtain

$$56.85 \pm (2.447)(6.48)$$

$$\text{or } 56.85 \pm 15.86 \text{ or } 40.99 \text{ to } 72.71.$$

Hence the desired 95% confidence interval for the true value of Y , i.e.,

$\mu_{Y,X}$ corresponding to $X=3.0$ is (40.99, 72.71).

question 3(c)